INTERIOR ESTIMATES FOR THE MONGE-AMPÈRE TYPED FOURTH ORDER EQUATIONS

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Abstract. In this paper, we first use the partial Legendre transform to study the interior regularity of a class of fourth order equations of Monge-Ampère type in dimension two. As an application, we obtain a new proof of the Bernstein theorem without using Caffarelli-Gutiérrez's estimate, including the Chern conjecture on the affine maximal surfaces. In higher dimensions, we also obtain the interior regularity in terms of the integral bounds on the second derivatives and the inverse of the determinant.

1. Introduction

We study the regularity of the following fourth order equations of Monge-Ampère type

$$U^{ij}w_{ij} = 0, \tag{1.1}$$

where (U^{ij}) is the cofactor matrix of D^2u of an unknown uniformly convex function, and

$$w = \begin{cases} [\det D^2u]^{-(1-\theta)}, & \theta \geq 0, \quad \theta \neq 1, \\ \log\det D^2u, & \theta = 1. \end{cases}$$

When $\theta = \frac{1}{n+2}$, it is the *affine maximal surface equation* in affine geometry [Ch]. When $\theta = 0$, it is *Abreu's equation* arise from the problem of extremal metrics on toric manifolds in Kähler geometry [Ab], and is equivalent to

$$\sum_{i,j} \frac{\partial^2 u^{ij}}{\partial x_i \partial x_j} = 0,$$

where (u^{ij}) is the inverse matrix of D^2u. The regularity of (1.1) has been extensively studied before [TW1, TW2, TW3, D1, Z1, Z2, CHLS, Le1, Le2]. This equation is usually treated as a system of a Monge-Ampère equation and a linearized Monge-Ampère equation. Therefore, in the former works, its regularity relies heavily on Caffarelli-Gutiérrez's deep result on the interior regularity of the linearized Monge-Ampère equation [CG], which was later extended by

2020 *Mathematics Subject Classification.* 35J30, 35G20, 35J96, 35B45, 35B65.

Key words and phrases. Monge-Ampère equation, linearized Monge-Ampère equation, partial Legendre transform.

*This research is partially supported by National Key R&D Program of China SQ2020YFA0712800 and NSFC grants 11822101.

[LS, GN1, GN2] to the the boundary and higher order estimates. In this paper, we first use partial Legendre transform to give a new proof of the interior estimate of (1.1) in dimension two. We will mainly concentrate on the case $\theta \in [0,1]$ due to the interesting geometric background.

Theorem 1.1. *Assume $n = 2$ and $\theta \in [0,1]$. Let $\Omega \subset \mathbb{R}^2$ be a convex domain and u be a uniformly convex solution to equation* (1.1) *on Ω satisfying*

$$0 < \lambda < \det D^2 u \leq \Lambda. \tag{1.2}$$

Then for any $\Omega' \Subset \Omega$, there exists a constant $C > 0$ depending on $\sup_\Omega |u|$, λ, Λ, θ and $dist(\Omega', \partial\Omega)$, such that

$$\|u\|_{C^{4,\alpha}(\Omega')} \leq C.$$

The partial Legendre transform to the fourth order equation was first used in [LZ] where the authors deal with the second boundary value problem. After the partial Legendre transform, equation (1.1) becomes a quasi-linear second order equation (2.2) for the determinant. The main ingredient in the proof of the above theorem is an interior integral estimate of the gradient(Theorem 3.1). When $\theta \in \left[0, \frac{1}{4}\right]$, condition (1.2) holds by the determinant estimates and arguments of the strict convexity [TW1, Z1]. Then we obtain a new proof of the following Bernstein theorem without using Caffarelli-Gutiérrez's theory.

Theorem 1.2. *Assume $n = 2$ and $0 \leq \theta \leq \frac{1}{4}$. Let u be an entire uniformly convex solution to* (1.1)*. Then u is a quadratic polynomial.*

In higher dimensions, the interior regularity and the Bernstein theorem are still widely open. In fact, according to the counterexample in [TW1] for the affine maximal surface equation, there may be no interior estimates if no further assumptions are made. Note that when condition (1.2) holds, one can use Caffarelli-Gutiérrez's theory to get the interior regularity. In this paper, we also investigate interior regularity in terms of the integral bounds on the second derivatives and the inverse of the determinant.

We first recall the regularity theory of the second order elliptic equation of divergence form

$$-D_j(a_{ij}(x)D_i u + b_j(x)u) + c_i(x)D_i u + d(x)u = f(x) \quad \text{in } \Omega,$$

where $\{a_{ij}(x)\}$ satisfy

$$a_{ij}(x)\xi_i\xi_j \geq \lambda(x)|\xi|^2 > 0, \quad \forall\, \xi \in \mathbb{R}^n \setminus \{0\}$$

and can be discontinuous. When $\{a_{ij}(x)\}$ is uniformly elliptic, a fundamental regularity theory was established by De Giorgi, Nash, Moser, etc. In [MS, Tr], the classical De Giorgi-Nash-Moser's theory was extended to degenerate linear elliptic equation with $\lambda(x)^{-1} \in L^p(\Omega)$ and $\Lambda(x) \in L^q(\Omega)$ where $\frac{1}{p} + \frac{1}{q} < \frac{2}{n}$, where $\Lambda(x)$ is the largest eigenvalue of $(a_{ij}(x))$. By the

divergence free property of U^{ij}, i.e., $\sum_i D_i U^{ij} = 0$, we can rewrite equation (1.1) as a divergence form

$$D_i(U^{ij} D_j w) = 0. \tag{1.3}$$

Inspired by the above results, we derive the following theorem.

Theorem 1.3. *Let $\Omega \subset \mathbb{R}^n$ be a convex domain and u be a uniformly convex solution to equation (1.1) with $\theta \in [0,1]$ on Ω. Assume that $p, q > 0$ satisfy $\frac{n}{p} + \frac{1}{q} < \frac{2}{n}$. Then for any $\Omega' \Subset \Omega$, there exists a constant C, depending only on p, q, θ, $\sup_\Omega |u|$, $\|u\|_{W^{2,p}(\Omega)}$, $\|(\det D^2 u)^{-1}\|_{L^q(\Omega)}$ and $dist(\Omega', \partial\Omega)$, such that*

$$\|u\|_{C^{4,\alpha}(\Omega')} \leq C.$$

A similar idea was used in the complex setting in [CC]. As an application, we obtain a Liouville type theorem in higher dimensions.

Corollary 1.4. *Let u be an entire uniformly convex solution to (1.1) with $\theta \in [0,1]$. Suppose there are $p, q, C > 0$ such that $\frac{n}{p} + \frac{1}{q} < \frac{2}{n}$ and*

$$\int_{B_R(0)} |D^2 u|^p + (\det D^2 u)^{-q}\, dx \leq C R^n\,, \quad \forall\, R > 0. \tag{1.4}$$

Then u is a quadratic polynomial.

The structure of the paper is as follows. In Section 2, we apply partial Legendre transform to (1.1) in dimension two to derive a new equation. The key interior gradient estimate(Theorem 3.1) for the new equation is established in Section 3. Then we prove Theorem 1.1 in Section 4 with this key estimate. Section 5 is devoted to some discussions on interior regularity in higher dimensions.

2. NEW EQUATION UNDER PARTIAL LEGENDRE TRANSFORM IN DIMENSION TWO

We first focus on dimension two case. Write $u(x) = u(x_1, x_2)$. The partial Legendre transform in the x_1-variable is

$$u^\star(\xi, \eta) = x_1 u_{x_1}(x_1, x_2) - u(x_1, x_2), \tag{2.1}$$

where

$$y = (\xi, \eta) = \mathcal{P}(x_1, x_2) := (u_{x_1}, x_2) \in \mathcal{P}(\Omega) =: \Omega^\star.$$

We have

$$\frac{\partial(\xi, \eta)}{\partial(x_1, x_2)} = \begin{pmatrix} u_{x_1x_1} & u_{x_1x_2} \\ 0 & 1 \end{pmatrix}, \quad \text{and} \quad \frac{\partial(x_1, x_2)}{\partial(\xi, \eta)} = \begin{pmatrix} \frac{1}{u_{x_1x_1}} & -\frac{u_{x_1x_2}}{u_{x_1x_1}} \\ 0 & 1 \end{pmatrix}.$$

Hence,

$$u^\star_\xi = x_1,\ u^\star_{\xi\xi} = \frac{1}{u_{x_1x_1}},\ u^\star_\eta = -u_{x_2},\ u^\star_{\eta\eta} = -\frac{\det D^2u}{u_{x_1x_1}},\ u^\star_{\xi\eta} = -\frac{u_{x_1x_2}}{u_{x_1x_1}}.$$

The partial Legendre transform has been used widely in the study of Monge-Ampère equation [DS, F, GP, LS, Li]. Here we apply it to equation (1.1).

In order to derive the equation under partial Legendre transform, we consider the associated functionals of (1.1)

$$A_\theta(u) = \begin{cases} \int_\Omega [\det D^2u]^\theta\, dx, & \theta > 0,\quad \theta \neq 1, \\ \int_\Omega \log\det D^2u\, dx, & \theta = 0, \\ \int_\Omega \det D^2u \log\det D^2u\, dx, & \theta = 1. \end{cases}$$

The case of $\theta = 0$ is essentially included in [LZ].

Proposition 2.1. *Let u be a uniformly convex solution to* (1.1) *in Ω. Then in $\Omega^\star = \mathcal{P}(\Omega)$, its partial Legendre transform $u^\star$ satisfies*

$$w^\star w^\star_{\xi\xi} + w^\star_{\eta\eta} + (\theta-1)w^{\star 2}_\xi + \frac{\theta-2}{w^\star} w^{\star 2}_\eta = 0, \tag{2.2}$$

Here $w^\star = -\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}}$.

Proof. By

$$\det D^2u = -\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}},\quad dxdy = u^\star_{\xi\xi}\, d\xi d\eta,$$

we have

$$\begin{aligned} A_\theta(u) &= \int_{\Omega^\star} \left(-\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}}\right)^\theta u^\star_{\xi\xi}\, d\xi d\eta \\ &= \int_{\Omega^\star} (-u^\star_{\eta\eta})^\theta {u^\star_{\xi\xi}}^{1-\theta}\, d\xi d\eta =: A^\star_\theta(u^\star),\ \theta \in (0,1); \\ A_0(u) &= \int_{\Omega^\star} \log\left(-\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}}\right) u^\star_{\xi\xi}\, d\xi d\eta =: A^\star_0(u^\star); \\ A_1(u) &= \int_{\Omega^\star} \left(-\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}}\right) \log\left(-\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}}\right) u^\star_{\xi\xi}\, d\xi d\eta =: A^\star_1(u^\star). \end{aligned}$$

It suffices to derive the Euler-Lagrange equation of $A^\star_\theta$.

First, we consider $\theta \in (0,1)$. For $\varphi \in C^\infty_0(\Omega^\star)$, by integration by parts,

$$\left.\frac{dA^\star_\theta(u^\star + t\varphi)}{dt}\right|_{t=0} = \int_{\Omega^\star} [-\theta(-u^\star_{\eta\eta})^{\theta-1}{u^\star_{\xi\xi}}^{1-\theta}\varphi_{\eta\eta} + (1-\theta)(-u^\star_{\eta\eta})^\theta (u^\star_{\xi\xi})^{-\theta}\varphi_{\xi\xi}]\, d\xi d\eta$$

$$= \int_{\Omega^\star} \{-[\theta(-u^\star_{\eta\eta})^{\theta-1} {u^\star_{\xi\xi}}^{1-\theta}]_{\eta\eta} + [(1-\theta)(-u^\star_{\eta\eta})^{\theta}(u^\star_{\xi\xi})^{-\theta}]_{\xi\xi}\}\varphi \, d\xi d\eta. \tag{2.3}$$

Denote $w^\star = -\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}}$. Then the equation after transformation is

$$-\theta(w^{\star\theta-1})_{\eta\eta} + (1-\theta)(w^{\star\theta})_{\xi\xi} = 0.$$

After simplification, it is (2.2). Similarly, for $\varphi \in C_0^\infty(\Omega^*)$,

$$\begin{aligned}\left.\frac{dA_0^\star(u^\star + t\varphi)}{dt}\right|_{t=0} &= \int_{\Omega^\star} -\frac{u^\star_{\xi\xi}}{u^\star_{\eta\eta}}\left(-\frac{\varphi_{\eta\eta}u^\star_{\xi\xi} - u^\star_{\eta\eta}\varphi_{\xi\xi}}{{u^\star_{\xi\xi}}^2}\right)u^\star_{\xi\xi} + \log\left(-\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}}\right)\varphi_{\xi\xi}\, d\xi d\eta \\ &= \int_{\Omega^\star} \frac{\varphi_{\eta\eta}u^\star_{\xi\xi} - u^\star_{\eta\eta}\varphi_{\xi\xi}}{u^\star_{\eta\eta}} + \log\left(-\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}}\right)\varphi_{\xi\xi}\, d\xi d\eta.\end{aligned}$$

Then the equation after transformation is

$$-[(w^\star)^{-1}]_{\eta\eta} + (\log w^\star)_{\xi\xi} = 0.$$

After simplification, we obtain (2.2). Finally,

$$\begin{aligned}\left.\frac{dA_1^\star(u^\star + t\varphi)}{dt}\right|_{t=0} &= \int_{\Omega^\star} (1+\log w^\star)\left(-\frac{\varphi_{\eta\eta}u^\star_{\xi\xi} - u^\star_{\eta\eta}\varphi_{\xi\xi}}{{u^\star_{\xi\xi}}^2}\right)u^\star_{\xi\xi} + w^\star \log w^\star \varphi_{\xi\xi}\, d\xi d\eta \\ &= -\int_{\Omega^\star} (1+\log w^\star)(w^\star\varphi_{\xi\xi} - \varphi_{\eta\eta}) + w^\star \log w^\star \varphi_{\xi\xi}\, d\xi d\eta.\end{aligned}$$

Then the equation after transformation is

$$-w^\star_{\xi\xi} - (\log w^\star)_{\eta\eta} = 0,$$

which is equivalent to (2.2). □

3. The interior gradient estimate of (2.2)

For simplicity, we change notations in this section and write (2.2) as

$$uu_{xx} + u_{yy} = (1-\theta)u_x^2 + \frac{2-\theta}{u}u_y^2. \tag{3.1}$$

It is easy to see that it is a quasi-linear equation with the right hand side depending on the gradient. We prove the following interior gradient estimate.

Theorem 3.1. *Assume u is a solution to (3.1) with $\theta \in [0,1]$ on $B_R(0)$ and satisfies $0 < \lambda \le u \le \Lambda$. Then there exist $\alpha, C > 0$ depending on λ, Λ, R and θ, such that*

$$\int_{B_R(0)} |\nabla u|^3 (R^2 - x^2 - y^2)^\alpha \, dV \le C. \tag{3.2}$$

Proof. Let $w = v\phi\eta$, where

$$v = \sqrt{u_x^2 + u_y^2 + 1},\ \eta = (R^2 - x^2 - y^2)^\alpha,\ \alpha > 3$$

and $\phi = \phi(u)$ is a positive function of u to be determined later.

Direct calculation yields

$$uw_{xx} + w_{yy} = (uv_{xx} + v_{yy})\phi\eta + 2[uv_x(\phi\eta)_x + v_y(\phi\eta)_y] + [u(\phi\eta)_{xx} + (\phi\eta)_{yy}]v. \tag{3.3}$$

It is clear that

$$v_x = v^{-1}(u_x u_{xx} + u_y u_{xy}), \quad v_y = v^{-1}(u_x u_{xy} + u_y u_{yy}), \tag{3.4}$$

$$v_{xx} = -v^{-3}(u_x u_{xx} + u_y u_{xy})^2 + v^{-1}(u_{xx}^2 + u_x u_{xxx} + u_{xy}^2 + u_y u_{xxy}), \tag{3.5}$$

$$v_{yy} = -v^{-3}(u_x u_{xy} + u_y u_{yy})^2 + v^{-1}(u_{xy}^2 + u_x u_{xyy} + u_{yy}^2 + u_y u_{yyy}). \tag{3.6}$$

Differentiating the equation, we have

$$u_x u_{xx} + u u_{xxx} + u_{xyy} = 2(1-\theta)u_x u_{xx} - \frac{2-\theta}{u^2}u_x u_y^2 + \frac{2(2-\theta)}{u}u_y u_{xy}, \tag{3.7}$$

$$u_y u_{xx} + u u_{xxy} + u_{yyy} = 2(1-\theta)u_x u_{xy} - \frac{2-\theta}{u^2}u_y u_y^2 + \frac{2(2-\theta)}{u}u_y u_{yy}. \tag{3.8}$$

By Cauchy inequality, we have

$$(u_x u_{xx} + u_y u_{xy})^2 \le (v^2-1)(u_{xx}^2 + u_{xy}^2), \quad (u_x u_{xy} + u_y u_{yy})^2 \le (v^2-1)(u_{xy}^2 + u_{yy}^2). \tag{3.9}$$

Then by (3.4), (3.5), (3.6), (3.7), (3.8) and (3.9), the first term in (3.3)

$$\begin{aligned}
& uv_{xx} + v_{yy} \\
=& u[v^{-1}(u_{xx}^2 + u_{xy}^2) - v^{-3}(u_x u_{xx} + u_y u_{xy})^2] + v^{-1}u(u_x u_{xxx} + u_y u_{xxy}) \\
& +[v^{-1}(u_{xy}^2 + u_{yy}^2) - v^{-3}(u_x u_{xy} + u_y u_{yy})^2] + v^{-1}(u_x u_{xyy} + u_y u_{yyy})) \\
\ge& v^{-1}\left[2(1-\theta)u_x(u_x u_{xx} + u_y u_{xy}) + \frac{2(2-\theta)}{u}u_y(u_x u_{xy} + u_y u_{yy}) - (v^2-1)u_{xx}\right] \\
& -v^{-1}\frac{2-\theta}{u^2}(u_x^2 u_y^2 + u_y^4) + v^{-3}[u(u_{xx}^2 + u_{xy}^2) + u_{xy}^2 + u_{yy}^2] \\
=& 2(1-\theta)u_x v_x + \frac{2(2-\theta)}{u}u_y v_y - \frac{2-\theta}{u^2}\frac{v^2-1}{v}u_y^2 - \frac{v^2-1}{v}u_{xx} \\
& +v^{-3}[u(u_{xx}^2 + u_{xy}^2) + u_{xy}^2 + u_{yy}^2)] \\
\ge& 2(1-\theta)u_x v_x + \frac{2(2-\theta)}{u}u_y v_y - \frac{2-\theta}{u^2}\frac{v^2-1}{v}u_y^2 - vu_{xx} + v^{-1}u_{xx} + v^{-3}uu_{xx}^2
\end{aligned}$$

$$\geq 2(1-\theta)u_xv_x + \frac{2(2-\theta)}{u}u_yv_y - \frac{2-\theta}{u^2}\frac{v^2-1}{v}u_y^2 - vu_{xx} - \frac{1}{2}\frac{v}{\sqrt{2u}} - v^{-3}uu_{xx}^2 + v^{-3}uu_{xx}^2$$

$$\geq 2(1-\theta)u_xv_x + \frac{2(2-\theta)}{u}u_yv_y - \frac{2-\theta}{u^2}\frac{v^2-1}{v}u_y^2 - vu_{xx} - Cv,$$

where $C > 0$ depends on λ. By integration by parts,

$$\int_{B_R(0)} (uv_{xx} + v_{yy})\phi\eta\, dV$$

$$\geq -\int_{B_R(0)} 2(1-\theta)[u_{xx}v\phi\eta + u_xv(\phi\eta)_x]\, dV$$

$$-\int_{B_R(0)} 2(2-\theta)\left[\frac{u_{yy}}{u}v\phi\eta - \frac{u_y^2}{u^2}v\phi\eta + \frac{1}{u}u_yv(\phi\eta)_y\right] dV$$

$$-\int_{B_R(0)} \left(\frac{2-\theta}{u^2}\frac{v^2-1}{v}u_y^2\phi\eta + vu_{xx}\phi\eta + C\phi\eta v\right) dV$$

$$= \int_{B_R(0)} u_{xx}v\phi\eta\, dV - \int_{B_R(0)} 2(2-\theta)\frac{1}{u}\left[(1-\theta)u_x^2 + \frac{2-\theta}{u}u_y^2\right] v\phi\eta\, dV$$

$$-\int_{B_R(0)} \left[2(1-\theta)u_xv(\phi\eta)_x + 2(2-\theta)\left(\frac{u_yv(\phi\eta)_y}{u} - \frac{u_y^2v\phi\eta}{u^2}\right) + \frac{2-\theta}{u^2}\left(v - \frac{1}{v}\right)u_y^2\phi\eta + C\phi\eta v\right] dV$$

$$= \int_{B_R(0)} u_{xx}v\phi\eta\, dV - \int_{B_R(0)} \left[2(1-\theta)\phi' + \frac{2(2-\theta)(1-\theta)}{u}\phi\right] u_x^2v\eta\, dV$$

$$-\int_{B_R(0)} \left[2(2-\theta)\frac{\phi'}{u} + (2-\theta)(3-2\theta)\frac{1}{u^2}\phi\right] u_y^2v\eta\, dV$$

$$-\int_{B_R(0)} \left[2(1-\theta)u_xv\phi\eta_x + \frac{2(2-\theta)}{u}u_yv\phi\eta_y + C\phi\eta v\right] dV + \int_{B_R(0)} \frac{2-\theta}{u^2}v^{-1}u_y^2\phi\eta\, dV$$

$$\geq \int_{B_R(0)} u_{xx}v\phi\eta\, dV - \int_{B_R(0)} \left[2(1-\theta)\phi' + \frac{2(2-\theta)(1-\theta)}{u}\phi\right] u_x^2v\eta\, dV$$

$$-\int_{B_R(0)} \left[2(2-\theta)\frac{\phi'}{u} + (2-\theta)(3-2\theta)\frac{1}{u^2}\phi\right] u_y^2v\eta\, dV$$

$$-C\left[\int_{B_R(0)} |\nabla u|^2(R^2-x^2-y^2)^{\alpha-1}\, dV + \int_{B_R(0)} |\nabla u|(R^2-x^2-y^2)^{\alpha}\, dV + 1\right].$$

Here $C > 0$ depends on λ, Λ, R and θ. The third term in (3.3)

$$[u(\phi\eta)_{xx} + (\phi\eta)_{yy}]v$$

$$
\begin{aligned}
&= v\left[\phi''(u)(uu_x^2+u_y^2)\eta+\phi'(u)(uu_{xx}+u_{yy})\eta+2\phi'(u)(uu_x\eta_x+u_y\eta_y)+\phi\cdot(u\eta_{xx}+\eta_{yy})\right]\\
&= v\Bigg[\phi''(u)(uu_x^2+u_y^2)\eta+\phi'(u)\left((1-\theta)u_x^2+\frac{2-\theta}{u}u_y^2\right)\eta+2\phi'(u)(uu_x\eta_x+u_y\eta_y)\\
&\quad +\phi\cdot(u\eta_{xx}+\eta_{yy})\Bigg]\\
&\le \left[(\phi''u+(1-\theta)\phi')u_x^2+\left(\phi''+\phi'\frac{2-\theta}{u}\right)u_y^2\right]v\eta\\
&\quad +C\left[|\nabla u|^2(R^2-x^2-y^2)^{\alpha-1}+|\nabla u|(R^2-x^2-y^2)^{\alpha-2}\right].
\end{aligned}
$$

Here $C>0$ depends on λ, Λ, R and θ. By integration by parts, the second term in (3.3)

$$
\begin{aligned}
&2\int_{B_R(0)} uv_x(\phi\eta)_x+v_y(\phi\eta)_y\,dV\\
&=-2\int_{B_R(0)} u_x(\phi\eta)_x v\,dV-2\int_{B_R(0)}[u(\phi\eta)_{xx}+(\phi\eta)_{yy}]v\,dV. \qquad (3.10)
\end{aligned}
$$

Hence

$$
\begin{aligned}
&\int_{B_R(0)} uw_{xx}+w_{yy}\,dV\\
&=\int_{B_R(0)}(uv_{xx}+v_{yy})\phi\eta\,dV-2\int_{B_R(0)}u_x(\phi\eta)_x v\,dV-\int_{B_R(0)}[u(\phi\eta)_{xx}+(\phi\eta)_{yy}]v\,dV\\
&\ge\int_{B_R(0)}u_{xx}v\phi\eta\,dV+\int_{B_R(0)}[(-5+3\theta)\phi'-\frac{2(2-\theta)(1-\theta)}{u}\phi-\phi''u]u_x^2v\eta\,dV\\
&\quad+\int_{B_R(0)}[-3(2-\theta)\frac{\phi'}{u}-(2-\theta)(3-2\theta)\frac{\phi}{u^2}-\phi'']u_y^2v\eta\,dV\\
&\quad -C\left[\int_{B_R(0)}|\nabla u|^2(R^2-x^2-y^2)^{\alpha-1}\,dV+\int_{B_R(0)}|\nabla u|(R^2-x^2-y^2)^{\alpha-2}\,dV+1\right]. \qquad (3.11)
\end{aligned}
$$

Note that the left hand side term

$$
\int_{B_R(0)} uw_{xx}+w_{yy}\,dV=\int_{B_R(0)}u_{xx}v\phi\eta\,dV. \qquad (3.12)
$$

Now we choose

$$
\phi(u)=Au^{\theta-2}-\frac{u}{2\theta^2-9\theta+9}
$$

with

$$
A\ge\frac{\Lambda^{3-\theta}}{2\theta^2-9\theta+9}+1.
$$

Then it is clear that $\phi(u) > 0$. Furthermore, since $\theta \in [0,1]$, we have

$$(-5+3\theta)\phi' - \frac{2(2-\theta)(1-\theta)}{u}\phi - \phi'' u = 1$$

and

$$-3(2-\theta)\frac{\phi'}{u} - (2-\theta)(3-2\theta)\frac{\phi}{u^2} - \phi'' = \frac{2(2-\theta)(3-\theta)}{(2\theta^2-9\theta+9)u} \geq C_0 > 0.$$

Combining them with (3.11) and (3.12), we obtain

$$\begin{aligned}
&\int_{B_R(0)} |\nabla u|^3 (R^2 - x^2 - y^2)^\alpha \, dV \\
\leq & C_1 \int_{B_R(0)} |\nabla u|^2 (R^2 - x^2 - y^2)^{\alpha-1} \, dV + C_2 \int_{B_R(0)} |\nabla u| (R^2 - x^2 - y^2)^{\alpha-2} \, dV + C_3 \\
\leq & C_1' \left(\int_{B_R(0)} |\nabla u|^3 (R^2 - x^2 - y^2)^\alpha \, dV \right)^{\frac{2}{3}} + C_2' \left(\int_{B_R(0)} |\nabla u|^3 (R^2 - x^2 - y^2)^\alpha \, dV \right)^{\frac{1}{3}} + C_3.
\end{aligned}$$

Hence, (3.2) follows. □

4. The interior regularity in dimension two

In order to use the partial Legendre transform, we first recall the modulus of convexity. For a convex function on $\mathbb{R}^n$, the *modulus of convexity* m_u of u is defined by

$$m_u(t) = \inf\{u(x) - \ell_z(x) : |x - z| > t\}, \tag{4.1}$$

where $t > 0$ and ℓ_z is the supporting function of u at z. For a strictly convex function, m_u must be a positive function. A result of Heinz [H] implies that in two dimensions, when $\det D^2 u \geq \lambda > 0$, there exists $C > 0$ depending on λ such that $m_u \geq C > 0$. Now for the partial Legendre transform (2.1), we consider the mapping

$$(\xi, \eta) = \mathcal{P}(x, y) = (u_x, y) : B_R(0) \to \mathbb{R}^2. \tag{4.2}$$

The following important property is revealed in [Li].

Lemma 4.1. [Li, Lemma 2.1] *There exists a constant $\delta > 0$ depending on the modulus of convexity m_u defined in* (4.1)*, such that $B_\delta(0) \subset \mathcal{P}(B_R(0))$.*

Proof of Theorem 1.1. For any $p \in \Omega$, let $R = \frac{dist(p, \partial\Omega)}{2}$. Without loss of generality, we assume $\mathcal{P}(p) = 0$. It is clear that $\sup_{B_R(p)} |Du| \leq C$ for some constant C depending on R and $\sup_\Omega |u|$.

By Lemma 4.1, there exists $\delta > 0$ such that $B_\delta(0) \subset \mathcal{P}(B_R(p))$. According to Proposition 2.1, $u^\star$ satisfies (2.2) in $B_\delta(0)$ with

$$0 < \lambda \leq w^\star = -\frac{u^\star_{\eta\eta}}{u^\star_{\xi\xi}} \leq \Lambda.$$

By Theorem 3.1,

$$\|w^\star\|_{W^{1,3}(B_{\frac{7\delta}{8}}(0))} \leq C.$$

Note that $n = 2$. By Sobolev theorem, we have the C^α estimate of $w^\star$. And by the interior $W^{2,p}$-estimate of the uniformly elliptic equation (2.2), we have the

$$\|w^\star\|_{W^{2,\frac{3}{2}}(B_{\frac{3\delta}{4}}(0))} \leq C,$$

which implies $W^{1,6}$-estimate of $w^\star$. Again by the interior $W^{2,p}$-estimate of the uniformly elliptic equation (2.2), we have the

$$\|w^\star\|_{W^{2,3}(B_{\frac{\delta}{2}}(0))} \leq C,$$

which implies $C^{1,\alpha}$ estimate of $w^\star$. Then by Shaulder estimate of (2.2), we have

$$\|w^\star\|_{C^{2,\alpha}(B_{\frac{\delta}{4}}(0))} \leq C$$

and all the higher order estimates of $u^\star$. Transforming back by partial Legendre transform, we obtain the interior estimates of u. □

5. The interior regularity in higher dimensions

In this section, we will prove Theorem 1.3. In view of [CG, Ca], it suffices to get the interior estimates on the upper and lower bound of $\det D^2 u$.

Recall that (1.1) is equivalent to

$$D_i(U^{ij} D_j w) = 0. \tag{5.1}$$

Since (U^{ij}) is a symmetric positive definite matrix, there exists a constant $c_n > 0$,

$$\frac{\det D^2 u}{c_n \cdot \Delta u} I \leq U^{ij} \leq c_n (\Delta u)^{n-1} I. \tag{5.2}$$

Denote $a_{ij} = U^{ij}$, $\lambda = \Delta u$ and $d = \det D^2 u$. We will prove the interior regularity under the assumption of $\lambda \in L^p(B_1)$ and $d^{-1} \in L^q(B_1)$.

First, we use the standard Moser's iteration to derive the following lemma for degenerate linear elliptic equation.

Lemma 5.1. *Assume $\{a_{ij}(x)\}$ satisfies*

$$\frac{d}{\lambda(x)} \leq a_{ij}(x) \leq \lambda(x)^{n-1} \quad \text{in } B_1, \tag{5.3}$$

where $\lambda(x) \in L^p(B_1)$, $d^{-1} \in L^q(B_1)$. Assume $c(x), f(x) \in L^{p_0}(B_1)$. Let $u \in W^{1,p}(B_1)$ be a subsolution in the following sense:

$$\int_{B_1} a_{ij} D_i u D_j \varphi + cu\varphi \, dx \le \int_{B_1} f\varphi \, dx \quad \text{for any } \varphi \in W_0^{1,p}(B_1) \text{ and } \varphi \ge 0. \tag{5.4}$$

Suppose p, q and p_0 satisfy $\frac{n}{p} + \frac{1}{q} < \frac{2}{n}$ and $p_0 \ge \frac{p}{n-1}$. Then

$$\sup_{B_{1/2}} u \le C \left(\|u^+\|_{L^1(B_1)} + \|f\|_{L^{p_0}(B_1)} \right),$$

where C depends only on p, $\|\lambda\|_{L^p(B_1)}$, $\|d^{-1}\|_{L^q(B_1)}$, $\|c\|_{L^{p_0}(B_1)}$.

Proof. For some $k > 0$ and $m > 0$, set $\overline{u} = u^+ + k$ and

$$\overline{u}_m = \begin{cases} \overline{u}, & u < m, \\ k + m, & u \ge m. \end{cases}$$

Let $\eta \in C_0^\infty(B_1)$, we choose the test function

$$\varphi = \eta^2 (\overline{u}_m^\beta \overline{u} - k^{\beta+1}) \in W_0^{1,p}(B_1)$$

for some $\beta \ge 0$ to be determined later. Substituting φ into (5.4), we have

$$\begin{aligned} &\beta \int_{B_1} a_{ij} D_i \overline{u} D_j \overline{u}_m \overline{u}_m^{\beta-1} \overline{u} \eta^2 \, dx + \int_{B_1} a_{ij} D_i \overline{u} D_j \overline{u} \, \overline{u}_m^\beta \eta^2 \, dx \\ &\qquad \le -2 \int_{B_1} a_{ij} D_i \overline{u} D_j \eta \overline{u}_m^\beta \overline{u} \eta \, dx + \int_{B_1} \left(|c| \eta^2 \overline{u}_m^\beta \overline{u}^2 + |f| \eta^2 \overline{u}_m^\beta \overline{u} \right) dx. \end{aligned}$$

Note that $D\overline{u} = D\overline{u}_m$ in $\{u < m\}$ and $D\overline{u}_m = 0$ in $\{u \ge m\}$. By Cauchy inequality, we have

$$\begin{aligned} &\beta \int_{B_1} a_{ij} D_i \overline{u}_m D_j \overline{u}_m \overline{u}_m^{\beta-1} \overline{u} \eta^2 \, dx + \int_{B_1} a_{ij} D_i \overline{u} D_j \overline{u} \, \overline{u}_m^\beta \eta^2 \, dx \\ &\le \frac{1}{2} \int_{B_1} a_{ij} D_i \overline{u} D_j \overline{u} \, \overline{u}_m^\beta \eta^2 \, dx + 4 \int_{B_1} a_{ij} D_i \eta D_j \eta \overline{u}_m^\beta \overline{u}^2 \, dx + \int_{B_1} \left(|c| \eta^2 \overline{u}_m^\beta \overline{u}^2 + |f| \eta^2 \overline{u}_m^\beta \overline{u} \right) dx. \end{aligned}$$

Then by (5.3)

$$\begin{aligned} &\beta \int_{B_1} \frac{d}{\lambda(x)} |D\overline{u}_m|^2 \overline{u}_m^\beta \eta^2 \, dx + \frac{1}{2} \int_{B_1} \frac{d}{\lambda(x)} |D\overline{u}|^2 \overline{u}_m^\beta \eta^2 \, dx \\ &\le \beta \int_{B_1} a_{ij} D_i \overline{u}_m D_j \overline{u}_m \overline{u}_m^{\beta-1} \overline{u} \eta^2 \, dx + \frac{1}{2} \int_{B_1} a_{ij} D_i \overline{u} D_j \overline{u} \, \overline{u}_m^\beta \eta^2 \, dx \\ &\le 4 \int_{B_1} \lambda(x)^{n-1} |D\eta|^2 \overline{u}_m^\beta \overline{u}^2 \, dx + \int_{B_1} c_0 \eta^2 \overline{u}_m^\beta \overline{u}^2 \, dx, \end{aligned} \tag{5.5}$$

where

$$c_0 = |c| + \frac{|f|}{k}.$$

Choose $k = \|f\|_{L^{p_0}(B_1)}$ if f is not identically 0. Otherwise choose arbitrary $k > 0$ and let $k \to 0^+$. Let $w = \overline{u}_m^{\frac{\beta}{2}}\overline{u}$. There holds

$$|Dw|^2 = \left|\frac{\beta}{2}\overline{u}_m^{\frac{\beta}{2}-1}\overline{u}_m D\overline{u}_m + \overline{u}_m^{\frac{\beta}{2}} D\overline{u}\right|^2 \leq (1+\beta)(\beta\overline{u}_m^{\beta}|D\overline{u}_m|^2 + \overline{u}_m^{\beta}|D\overline{u}|^2).$$

Therefore by (5.5)

$$\int_{B_1} \frac{d}{\lambda(x)}|Dw|^2\eta^2\,dx \leq 8(1+\beta)\int_{B_1}\lambda(x)^{n-1}|D\eta|^2w^2\,dx + 2(1+\beta)\int_{B_1} c_0w^2\eta^2\,dx.$$

By

$$|D(w\eta)|^2 \leq 2|Dw|^2\eta^2 + 2|D\eta|^2w^2,$$

we have

$$\begin{aligned}\int_{B_1} \frac{d}{\lambda(x)}|D(w\eta)|^2\,dx &\leq \int_{B_1}\left(\frac{2d}{\lambda(x)} + 16(1+\beta)\lambda(x)^{n-1}\right)|D\eta|^2w^2\,dx + 4(1+\beta)\int_{B_1} c_0w^2\eta^2\,dx \\ &\leq 18(1+\beta)\int_{B_1}\lambda(x)^{n-1}|D\eta|^2w^2\,dx + 4(1+\beta)\int_{B_1} c_0w^2\eta^2\,dx. \qquad (5.6)\end{aligned}$$

Next, we deal with the $\lambda(x)$ in the above estimate. By the assumptions $\lambda \in L^p(B_1)$ and $d^{-1} \in L^q(B_1)$, we have $\lambda d^{-1} \in L^{\frac{pq}{p+q}}(B_1)$. By Hölder inequality, we have

$$\|D(w\eta)\|^2_{L^{\frac{2pq}{pq+p+q}}(B_1)} \leq \|\lambda d^{-1}\|_{L^{\frac{pq}{p+q}}(B_1)} \cdot \int_{B_1}\frac{1}{\lambda(x)d^{-1}}\,|D(w\eta)|^2\,dx, \qquad (5.7)$$

$$\int_{B_1}\lambda(x)^{n-1}|D\eta|^2w^2\,dx \leq \|\lambda\|_{L^p(B_1)} \cdot \|wD\eta\|^2_{L^{\frac{2p}{p-n+1}}(B_1)}, \qquad (5.8)$$

$$\int_{B_1} c_0w^2\eta^2\,dx \leq \|c_0\|_{L^{p_0}(B_1)} \cdot \left(\int_{B_1}(\eta w)^{\frac{2p_0}{p_0-1}}\,dx\right)^{1-\frac{1}{p_0}}. \qquad (5.9)$$

Combining (5.6)-(5.9), we get

$$\begin{aligned}\|D(w\eta)\|_{L^{\frac{2pq}{pq+p+q}}(B_1)} &\leq C(1+\beta)^{\frac{1}{2}}\left(\|wD\eta\|_{L^{\frac{2p}{p-n+1}}(B_1)} + \|w\eta\|_{L^{\frac{2p_0}{p_0-1}}(B_1)}\right) \\ &\leq C(1+\beta)^{\frac{1}{2}}\left(\|wD\eta\|_{L^{\frac{2p}{p-n+1}}(B_1)} + \|w\eta\|_{L^{\frac{2p}{p-n+1}}(B_1)}\right).\end{aligned}$$

where C depending on $\|\lambda\|_{L^p(B_1)}$, $\|d^{-1}\|_{L^q(B_1)}$, $\|c\|_{L^{p_0}(B_1)}$. Here we used $p_0 \geq \frac{n-1}{p}$. By Sobolev inequality

$$\|w\eta\|_{L^\alpha(B_1)} \leq C(1+\beta)^{\frac{1}{2}}\left(\|wD\eta\|_{L^{\frac{2p}{p-n+1}}(B_1)} + \|w\eta\|_{L^{\frac{2p}{p-n+1}}(B_1)}\right),$$

where

$$\frac{1}{\alpha} = \frac{pq+p+q}{2pq} - \frac{1}{n}. \qquad (5.10)$$

Now for any $0 < r < R \leq 1$, we choose the cutoff function $\eta \in C_0^\infty(B_R)$ with the property

$$0 \leq \eta \leq 1, \quad \eta \equiv 1 \text{ in } B_r \quad \text{and} \quad |D\eta| \leq \frac{2}{R-r}.$$

Then we obtain

$$\|w\|_{L^\alpha(B_r)} \leq \frac{C(1+\beta)^{\frac{1}{2}}}{R-r} \|w\|_{L^{\frac{2p}{p-n+1}}(B_R)}.$$

By (5.10) and the assumption $\frac{n}{p} + \frac{1}{q} < \frac{2}{n}$, we have $\alpha > \frac{2p}{p-n+1}$. We can do the iteration as follows.

Recalling the definition of w, we have

$$\|\overline{u}_m^{\frac{\beta}{2}} \overline{u}\|_{L^\alpha(B_r)} \leq \frac{C(1+\beta)^{\frac{1}{2}}}{R-r} \|\overline{u}_m^{\frac{\beta}{2}} \overline{u}\|_{L^{\frac{2p}{p-n+1}}(B_R)}.$$

Set $\gamma = \beta + 2 \geq 2$. By $\overline{u}_m \leq \overline{u}$, we obtain

$$\|\overline{u}_m^{\frac{\gamma}{2}}\|_{L^\alpha(B_r)} \leq \frac{C\gamma^{\frac{1}{2}}}{R-r} \|\overline{u}^{\frac{\gamma}{2}}\|_{L^{\frac{2p}{p-n+1}}(B_R)}.$$

Letting $m \to \infty$, we get

$$\|\overline{u}^{\frac{\gamma}{2}}\|_{L^\alpha(B_r)} \leq \frac{C\gamma^{\frac{1}{2}}}{R-r} \|\overline{u}^{\frac{\gamma}{2}}\|_{L^{\frac{2p}{p-n+1}}(B_R)},$$

i.e.,

$$\|\overline{u}\|_{L^{\frac{\gamma}{2}\alpha}(B_r)} \leq \frac{(C\gamma)^{\frac{1}{\gamma}}}{(R-r)^{\frac{2}{\gamma}}} \|\overline{u}\|_{L^{\frac{\gamma}{2}\frac{2p}{p-n+1}}(B_R)}.$$

Denote $\chi = \alpha \cdot \frac{(p-n+1)}{2p} > 1$. Then

$$\|\overline{u}\|_{L^{\frac{\gamma p}{p-n+1}\chi}(B_r)} \leq \frac{(C\gamma)^{\frac{1}{\gamma}}}{(R-r)^{\frac{2}{\gamma}}} \|\overline{u}\|_{L^{\frac{\gamma p}{p-n+1}}(B_R)}. \tag{5.11}$$

We iterate (5.11) to get the desired estimate. Set

$$\gamma_i = 2\chi^i \quad \text{and} \quad R_i = r + \frac{R-r}{2^i}, \ i = 0, 1, 2, \cdots,$$

i.e.,

$$\gamma_i = \chi\gamma_{i-1} \text{ and } R_{i-1} - R_i = \frac{R-r}{2^i}, \ i = 1, 2, \cdots.$$

By (5.11),

$$\|\overline{u}\|_{L^{\frac{2p}{p-n+1}\chi^{i+1}}(B_{R_{i+1}})} \leq C^{\sum_{j=0}^i \frac{1}{\gamma_j}} \cdot \prod_{j=0}^i \gamma_j^{\frac{1}{\gamma_j}} \cdot 4^{\sum_{j=0}^i \frac{j}{\gamma_j}} \frac{1}{(R-r)^{\sum_{j=0}^i \frac{2}{\gamma_j}}} \cdot \|\overline{u}\|_{L^{\frac{2p}{p-n+1}}(B_R)}.$$

Letting $i \to \infty$, by Young's inequality, we have

$$\begin{aligned}\|\overline{u}\|_{L^\infty(B_r)} &\le \frac{C}{(R-r)^{\frac{\chi}{\chi-1}}}\|\overline{u}\|_{L^{\frac{2p}{p-n+1}}(B_R)} \\ &= \frac{C}{(R-r)^{\frac{\chi}{\chi-1}}}\|\overline{u}\|_{L^1(B_R)}^{\frac{p-n+1}{2p}} \cdot \|\overline{u}\|_{L^\infty(B_R)}^{\frac{p+n-1}{2p}} \\ &\le \frac{1}{2}\|\overline{u}\|_{L^\infty(B_R)} + \frac{C}{(R-r)^{\frac{\chi}{\chi-1}\cdot\frac{2p}{p-n+1}}}\|\overline{u}\|_{L^1(B_R)}.\end{aligned}$$

Set $f(t) = \|\overline{u}\|_{L^\infty(B_t)}$ for $t \in (0,1]$. Then for any $0 < r < R \le 1$

$$f(r) \le \frac{1}{2}f(R) + \frac{C}{(R-r)^{\frac{\chi}{\chi-1}\cdot\frac{2p}{p-n+1}}}\|\overline{u}\|_{L^1(B_1)}.$$

We apply the Lemma 5.2 below to get

$$f(r) \le \frac{C}{(R-r)^{\frac{\chi}{\chi-1}\cdot\frac{2p}{p-n+1}}}\|\overline{u}\|_{L^1(B_1)}.$$

The lemma follows by choosing $r = \frac{1}{2}$, $R = 1$. □

Lemma 5.2. [HL, Lemma 4.3] *Let $f(t) \ge 0$ be bounded in $[\tau_0, \tau_1]$ with $\tau_0 \ge 0$. Suppose for $\tau_0 \le t < s \le \tau_1$ we have*

$$f(t) \le \beta f(s) + \frac{A}{(s-t)^\alpha} + B$$

for some $\beta \in [0,1)$. Then for any $\tau_0 \le t < s \le \tau_1$ there holds

$$f(t) \le C(\alpha,\beta)\left\{\frac{A}{(s-t)^\alpha} + B\right\}.$$

Now we can use Lemma 5.1 to obtain the interior estimates.

Proof of Theorem 1.3. Firstly, we consider the case $0 \le \theta < 1$. We apply Lemma 5.1 to equation (5.1), which yields

$$\sup_{B_{1/2}} w \le C\|w\|_{L^1(B_1)} \le C\|w\|_{L^{\frac{q}{1-\theta}}(B_1)}.$$

Since $w = (\det D^2u)^{-(1-\theta)} \in L^{\frac{q}{1-\theta}}(B_1)$, we know

$$\sup_{B_{1/2}}\{(\det D^2u)^{-1}\} \le C.$$

For the upper bound of the determinant, we set $w = \frac{1}{v}$. Direct calculation yields

$$w_i = -\frac{v_i}{v^2},$$

$$w_{ij} = \frac{2v_iv_j}{v^3} - \frac{v_{ij}}{v^2}.$$

Then v satisfies

$$0 = U^{ij} w_{ij} = \frac{2}{v^3} U^{ij} v_i v_j - \frac{1}{v^2} U^{ij} v_{ij} \geq -\frac{1}{v^2} U^{ij} v_{ij},$$

i.e.,

$$-D_i(U^{ij} D_j v) \leq 0 \quad \text{in } B_1.$$

Similarly, we use Lemma 5.1 to obtain

$$\sup_{B_{1/2}} v \leq C\|v\|_{L^1(B_1)} \leq C\|v\|_{L^{\frac{p}{n(1-\theta)}}(B_1)},$$

which implies

$$\sup_{B_{1/2}} \det D^2 u \leq C.$$

Next, we consider the case $\theta = 1$. Write $w = \log d$, where $d = \det D^2 u \in L^{\frac{p}{n}}(B_1)$. Then we have

$$w_i = \frac{d_i}{d},$$

$$w_{ij} = -\frac{d_i d_j}{d^2} + \frac{d_{ij}}{d},$$

which yields

$$0 = U^{ij} w_{ij} = -\frac{1}{d^2} U^{ij} d_i d_j + \frac{1}{d} U^{ij} d_{ij} \leq \frac{1}{d} U^{ij} d_{ij},$$

i.e.,

$$-D_i(U^{ij} D_j d) \leq 0 \quad \text{in } B_1.$$

Similarly, write $w = -\log z$ with $z = (\det D^2 u)^{-1} \in L^q(B_1)$. We have

$$w_i = -\frac{z_i}{z},$$

$$w_{ij} = \frac{z_i z_j}{z^2} - \frac{z_{ij}}{z},$$

which means

$$0 = U^{ij} w_{ij} = \frac{1}{z^2} U^{ij} z_i z_j - \frac{1}{z} U^{ij} z_{ij} \geq -\frac{1}{z} U^{ij} z_{ij},$$

i.e.,

$$-D_i(U^{ij} D_j z) \leq 0 \quad \text{in } B_1.$$

Then we use Lemma 5.1 to obtain the bounds of $\det D^2 u$.

As we mentioned before, once we have the determinant estimates, all the interior estimates follows. □

Remark 5.3. *To get the upper bound of* $\det D^2u$*, we only need* $u \in W^{2,p}(B_1)$ *with* $p > \frac{n^2}{2}$*. In fact, by taking* $\det D^2u = v^{\frac{1}{1-\theta}}$ $(0 \le \theta < 1)$*, where* v *defined as the proof of Theorem 1.3, condition* (5.3) *in Lemma 5.1 becomes*

$$\frac{v^{\frac{1}{1-\theta}}}{\lambda(x)} \le a_{ij}(x) \le (\lambda(x))^{n-1}$$

Then the upper bound can be obtained by similar arguments as Lemma 5.1 without assumption on $(\det D^2u)^{-1}$*.*

Remark 5.4. *We can also consider the nonhomogeneous equation*

$$D_j(U^{ij}D_iw) = f \quad \text{in } B_1 \tag{5.12}$$

where $f \in L^{p_0}(B_1)$*. For the lower bound of* $\det D^2u$*, if we assume* $p_0 \ge \frac{p}{n-1}$*, we can apply Lemma 5.1 to equation* (5.12) *directly for all* $\theta \in [0,1)$*. However, for the upper bound of* $\det D^2u$*,* $v = \frac{1}{w}$ *satisfies*

$$-D_j(U^{ij}D_iv) \le fv^2 \quad \text{in } B_1. \tag{5.13}$$

Even we assume $p_0 = \infty$*, we only know* $fv \in L^{\frac{p}{n(1-\theta)}}(B_1)$*. Then we can only apply Lemma 5.1 to the equation* (5.13) *to get the upper bound of* $\det D^2u$ *when* $\frac{1}{n} \le \theta < 1$*. Hence, we have all the higher interior estimates for* $\theta \in [\frac{1}{n}, 1)$*. Note that the case* $\theta = 0$*(Abreu's equation) and* $\theta = \frac{1}{n+2}$*(affine maximal surface equation) are not included.*

Finally, we prove a Liouville type theorem.

Proof of Corollary 1.4. For u in any $B_R \subset \mathbb{R}^n$, define

$$u_R(x) = \frac{1}{R^2}u(Rx),$$

then we know $U_R^{ij}(w_R)_{ij} = 0$ in B_1. Since u satisfies (1.4), we know u_R satisfies

$$\int_{B_1} |D^2u_R|^p + (\det D^2u_R)^{-q}\,dx \le C.$$

Applying Theorem 1.3 to u_R, there exists a $C > 0$ independent of R, such that $\|u_R\|_{C^{4,\alpha}(B_{1/2})} \le C$. In particular, we know $\|D^3u_R\|_{L^\infty(B_{1/2})} \le C$. Hence

$$\|D^3u(Rx)\|_{L^\infty(B_{\frac{1}{2}})} \le \frac{C}{R},$$

i.e.,

$$\|D^3u(x)\|_{L^\infty(B_{\frac{R}{2}})} \le \frac{C}{R}.$$

Let $R \to +\infty$, we have $D^3u \equiv 0$, which means u is a quadratic function. □

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